

Multichannel scheduling and routing topologies for throughput-delay tradeoff

Joint frequency and time slot assignment

- NP-hard for general graphs
- Approximation algorithms with constant factor and logarithmic bounds
- Bounded-degree, minimum-radius spanning trees (BDMRST)

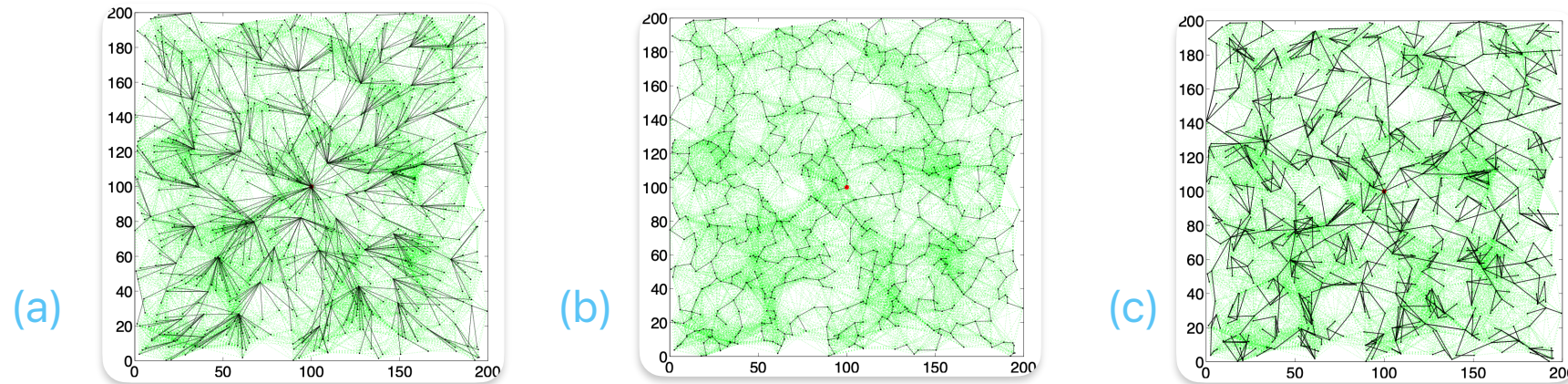
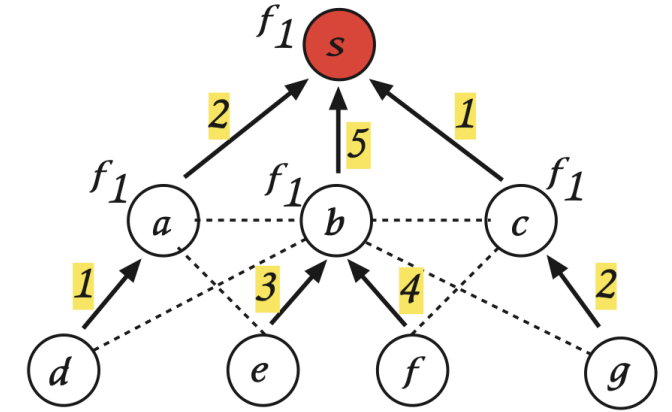
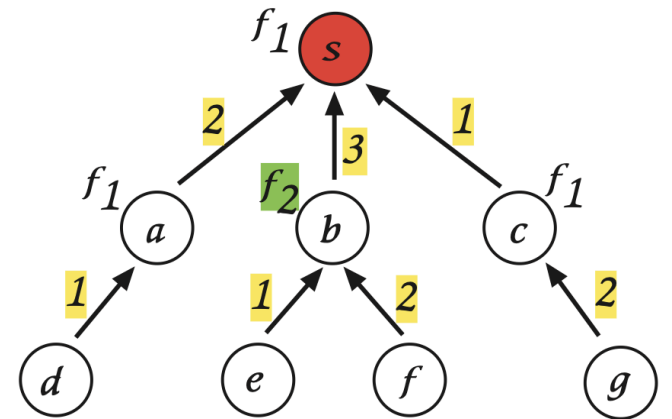


Fig 2. 800 nodes randomly deployed in a 200 x 200 region, with the sink at the center.
(a) Shortest-path tree: Minimum hop distances to the sink, but high node degrees,
(b) Minimum interference tree: Low node degrees, but large hop distances to the sink,
(c) BDMRST: Uniform node degrees, and uniform hop distances to the sink

Fig 1. Convergecast: Data flow from all nodes (a, ..., g) to sink s using a TDMA schedule. Arrows represent transmission directions, and numbers indicate time slots. A node transmits as per its receiver's frequency, e.g., d transmits over f_1 .



(Top) When transmissions occur over a single frequency f_1 , interfering links (dotted lines) come into play, and 5 time slots are needed for the sink to start receiving aggregated data from all the nodes in steady state.



(Bottom) When two frequencies f_1 and f_2 are used, interfering links disappear, and only 3 time slots are enough.